Negation-Limited Monotone Circuits for CLIQUE: A Simple Proof of Strong Lower Bounds

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Abstract

We revisit the monotone circuit complexity of the CLIQUE function. Razborov's classic method shows that any monotone circuit computing $\mathrm{CLIQUE}(n, k)$ requires size $n^{\Omega(\sqrt{k})}$. We prove that this lower bound remains valid even if the circuit is allowed to use up to $c \log n$ negation (NOT) gates for any constant c. The proof is a direct extension of Razborov's approximation framework: the sole effect of a NOT gate is to introduce an additional error term bounded by $|X| + |Y|$, where X and Y are the sets of positive and negative examples. Since Razborov's threshold $Q = |X| \cdot |Y|$ is exponentially larger than $|X| + |Y|$, a logarithmic number of negations cannot significantly affect the lower bound. This gives a simpler proof that handles a larger range of NOT gates ($c \log n$ vs. the previous $(1/6) \log \log n$ of Amano and Maruoka, 2005) by a cruder—but asymptotically sufficient—error analysis.

We also provide a hands-on verification of Razborov's method on a minimal instance ($n = 5, k = 3$) and an explicit Karchmer-Wigderson game for the same parameters, as pedagogical illustrations of the mechanisms behind the lower bounds.

1 Introduction

The question of whether negation gates can substantially reduce the monotone circuit complexity of a function is a classical topic in circuit complexity. Markov's theorem [3] shows that any Boolean function can be computed by a circuit with at most $\lceil \log_2(n + 1) \rceil$ NOT gates, but this does not bound the total number of AND/OR gates. Amano and Maruoka [2] demonstrated that for the NP-complete CLIQUE function, super-polynomial monotone size is required even when the circuit is allowed to use a very limited number of NOT gates: up to $(1/6) \log \log n$. Their proof combines Razborov's approximation method with a delicate induction on circuit layers to keep the approximation parameters (m, l) under control after each negation.

In this note we give a much simpler proof that handles a larger range of NOT gates: any circuit for $\mathrm{CLIQUE}(n, k)$ that contains at most $c \log n$ NOT gates must contain at least $n^{\Omega(\sqrt{k})}$ AND/OR gates. The argument rests on a trivial observation: the error introduced by a NOT gate under the Razborov approximation is at most the total number of critical inputs $|X| + |Y|$, which is negligible compared to the product $|X| \cdot |Y|$ that drives Razborov's lower bound. By sacrificing the fine-grained error control of Amano and Maruoka's induction, we obtain a cruder bound that nevertheless suffices for the asymptotic regime.

We complement this result with two pedagogical examples:

- A step-by-step manual verification of Razborov's approximator construction on a minimal CLIQUE instance ($n = 5, k = 3$).
- An explicit Karchmer-Wigderson game for the same instance, demonstrating the communication-complexity perspective on monotone depth.

These examples serve to ground the abstract theory and are *not* part of the asymptotic lower-bound proof; the parameters of the $n = 5$ instance do not lie in the regime where the main theorem applies.

2 Preliminaries

2.1 The CLIQUE function

Let n and k be integers, $k \leq n$. The function $\text{CLIQUE}(n,k) : \{0,1\}^{\binom{n}{2}} \to \{0,1\}$ outputs 1 iff the input graph (represented by its adjacency matrix) contains a clique of size k.

Following Razborov [1], we fix two families of inputs:

- X: all graphs consisting of exactly one k-clique and no other edges. $|X| = \binom{n}{k}$.
- Y: all complete $(k-1)$-partite graphs, i.e., graphs whose vertices can be colored with $k-1$ colors such that only edges between different color classes are present. $|Y| = (k-1)^n$.

Clearly, for $x \in X$, $\text{CLIQUE}(n,k)(x) = 1$ and for $y \in Y$, $\text{CLIQUE}(n,k)(y) = 0$.

2.2 Monotone circuits and approximations

A *monotone circuit* is a Boolean circuit using only AND and OR gates (no NOT gates). Razborov [1] introduced the method of *approximators* to prove lower bounds for monotone circuits. An (m,l)-approximator is a monotone DNF formula of the form

$$A = \bigvee_{t=1}^{r} \lceil X_t \rceil,$$

where $r \leq m$, each X_t is a subset of vertices with $|X_t| \leq l$, and $\lceil X_t \rceil$ denotes the indicator that the input graph contains a clique on X_t. Note that the empty disjunction ($r = 0$) is the constant-0 approximator, and taking $X_t = \emptyset$ yields the constant-1 approximator; both are legal (m,l)-approximators for any $m, l \geq 0$.

The approximation procedure replaces each gate of a monotone circuit by such an approximator in a bottom-up fashion:

- An input variable x_{ij} is represented by the approximator $\lceil \{i,j\} \rceil$.
- An AND gate $A_1 \wedge A_2$: compute all pairwise unions $X_s \cup Y_t$ of the clique sets of A_1 and A_2; drop those of size $> l$; reduce the number of sets to $\leq m$ using the Sunflower Lemma [5].
- An OR gate $A_1 \vee A_2$: take the union of the two clique families; if the number exceeds m, repeatedly find a sunflower and replace it by its core.

The key property is that the AND approximation only introduces *false negatives* (errors on X), while the OR approximation only introduces *false positives* (errors on Y). Let $E^+(A)$ denote the number of $x \in X$ with $A(x) = 0$ while the true circuit outputs 1, and $E^-(A)$ the number of $y \in Y$ with $A(y) = 1$ while the true output is 0. For a correct circuit we must finally have $E^+ = E^- = 0$.

Razborov proved that each monotone gate increases the total error $E^+ + E^-$ by at most some constant Δ (depending on the parameters l, m but independent of n). Since a correct circuit must have zero error, the number of AND/OR gates T must satisfy $T \cdot \Delta \geq Q$, where $Q = |X| \cdot |Y|$. Choosing $l = \Theta(\sqrt{k})$ yields $T \geq n^{\Omega(\sqrt{k})}$.

3 Negation gates

Now consider a circuit C computing $\text{CLIQUE}(n,k)$ that contains s NOT gates, in addition to T AND/OR gates. We analyze how the NOT gates affect the approximation framework.

A NOT gate receives a Boolean value g (produced by a monotone subcircuit) and outputs $\neg g$. In the approximation process we need to replace $\neg g$ by an (m, l)-approximator $\hat{g}$. The function $\neg g$ is anti-monotone, and its approximator will necessarily introduce some error. We bound this error by a trivial observation:

Lemma 3.1. *Let $h : X \cup Y \to \{0, 1\}$ be any Boolean function. There exists a constant approximator (either identically 0 or identically 1) whose total error $E^+ + E^-$ on $X \cup Y$ is at most $\frac{|X|+|Y|}{2}$.*

Proof. If $|\{h = 1\}| \leq |\{h = 0\}|$, take the constant 0 function; then $E^+ = |\{h = 1\}|$ and $E^- = 0$. Otherwise take constant 1, giving $E^- = |\{h = 0\}|$ and $E^+ = 0$. In either case the total error is $\min(|\{h = 1\}|, |\{h = 0\}|) \leq \frac{|X|+|Y|}{2}$. □

Lemma 3.2. *The constant functions 0 and 1 are legal (m, l)-approximators for any $m, l \geq 0$. Consequently, after replacing a NOT gate by a constant approximator, all subsequent AND/OR gates can be processed by the standard Razborov approximation rules, and each contributes at most Δ to the total error.*

Proof. The constant 0 is the empty disjunction ($r = 0 \leq m$). The constant 1 is obtained by taking $X_1 = \emptyset$ (since $|\emptyset| = 0 \leq l$, and every graph vacuously contains a clique on the empty set). Both satisfy the definition of an (m, l)-approximator. Since Razborov's error analysis for each AND/OR gate depends only on the fact that its inputs are (m, l)-approximators (not on their accuracy), the standard per-gate error bound Δ applies unchanged. □

3.1 Error accumulation with multiple NOT gates

The circuit consists of s NOT gates and T AND/OR gates. We apply the Razborov approximation to every monotone subcircuit (the parts between NOT gates). By Lemma 3.2, each AND/OR gate contributes at most Δ to the total error budget. By Lemma 3.1, each NOT gate introduces an extra error of at most $(|X|+|Y|)/2$.

Thus, the total error before the final output is bounded by

$$E_{\text{total}} \leq T\,\Delta + s \cdot \frac{|X| + |Y|}{2}.$$

For the circuit to correctly compute CLIQUE, the approximator at the output must satisfy $E^+ = E^- = 0$, i.e., $E_{\text{total}} = 0$. Since E_{total} is integer-valued, if the upper bound in the above inequality is strictly less than 1, then $E_{\text{total}} = 0$. More generally, Razborov's analysis establishes a threshold: for an approximator to separate X from Y correctly, at least Q/Δ AND/OR gates are required. In the presence of NOT gates, the condition becomes:

$$T\Delta + s\frac{|X| + |Y|}{2} \;\geq\; Q,$$

since otherwise the monotone part of the circuit would have insufficient "error budget" to achieve the separation. Rearranging,

$$T \;\geq\; \frac{Q - s\frac{|X|+|Y|}{2}}{\Delta}.$$

If $s\frac{|X|+|Y|}{2} \leq \frac{Q}{2}$, i.e., $s \leq \frac{Q}{|X|+|Y|}$, then

$$T \;\geq\; \frac{Q}{2\Delta} \;=\; n^{\Omega(\sqrt{k})}.$$

3.2 The magnitude of $Q/(|X|+|Y|)$

We have $|X| = \binom{n}{k}$ and $|Y| = (k-1)^n$. For $k = \Theta(n^{1/4})$ (the usual setting for Razborov's lower bound),

$$|X| = n^{\Theta(n^{1/4})}, \qquad |Y| = n^{\Theta(n^{3/4})},$$

and therefore

$$\frac{Q}{|X|+|Y|} = \frac{|X|\cdot|Y|}{|X|+|Y|} \geq \frac{1}{2}\min(|X|,|Y|) = n^{\Theta(n^{1/4})}.$$

This quantity is super-polynomially large. Hence any $s = O(\log n)$ (indeed any $s = o(\min(|X|,|Y|))$) satisfies the required inequality.

4 Main theorem

Theorem 4.1. *Let C be a Boolean circuit computing $\mathrm{CLIQUE}(n,k)$ that contains s NOT gates and T AND/OR gates. If $s = o\Big(\frac{|X|\cdot|Y|}{|X|+|Y|}\Big)$, then $T \geq n^{\Omega(\sqrt{k})}$. In particular, for any constant c, circuits with at most $c\log n$ NOT gates require super-polynomial AND/OR size.*

Proof. From Section 3, the number of AND/OR gates must satisfy

$$T \;\geq\; \frac{Q - s\,(|X|+|Y|)/2}{\Delta}.$$

Under the hypothesis on s, the numerator is $\Omega(Q)$, yielding $T = \Omega(Q/\Delta) = n^{\Omega(\sqrt{k})}$. □

5 Hands-on verification: Razborov's method on a minimal example

To provide a concrete illustration of the abstract framework, we work through the case $n = 5, k = 3$. *This is purely pedagogical; the $n = 5$ instance does not lie in the asymptotic regime $k = \Theta(n^{1/4})$ for which the main theorem yields a meaningful lower bound.*

The input graph has $\binom{5}{2} = 10$ edges. The positive set X consists of the $\binom{5}{3} = 10$ graphs containing exactly one triangle. The negative set Y consists of the 2-colorings of the vertices (complete bipartite graphs), of which there are $2^{5-1} = 16$ distinct edge structures, or 32 labeled colorings (fixing $c(1) = 0$).

5.1 Parameters and approximators

We set $l = 2$ (clique indicators of size at most 2) and $p = 2$ (sunflower petals), giving $m = l!\cdot(p-1)^l = 2\cdot 1^2 = 2$. Thus each approximator can contain at most two clique indicators, each on at most two vertices.

5.2 Approximating an AND gate

Consider $G_{123} = x_{12} \wedge x_{13} \wedge x_{23}$. The three inputs are the edge indicators $\lceil\{1,2\}\rceil, \lceil\{1,3\}\rceil, \lceil\{2,3\}\rceil$. Pairwise AND produces the union sets:

$$\lceil\{1,2\}\rceil \wedge \lceil\{1,3\}\rceil = \lceil\{1,2,3\}\rceil.$$

Since $|\{1,2,3\}| = 3 > l = 2$, this clique is dropped. The resulting clique set is empty, so the approximator is the constant 0.

Errors: The true gate G_{123} outputs 1 only on the triangle $\{1,2,3\}$. The constant-0 approximator misses this, producing one false negative ($E^+ = 1$) and no false positives ($E^- = 0$). This illustrates the AND rule: the approximator is $\leq$ the true function, only increasing E^+.

5.3 Approximating an OR gate

Now take $A_{123} = \lceil\{1,2\}\rceil \vee \lceil\{1,3\}\rceil \vee \lceil\{2,3\}\rceil$. Merging gives three cliques: $\{1,2\}, \{1,3\}, \{2,3\}$. The count $r = 3$ exceeds $m = 2$, triggering sunflower compression. The cliques $\{1,2\}$ and $\{1,3\}$ share the core $\{1\}$; they form a sunflower with 2 petals. Replacing them by the core yields the set $\{\{1\}, \{2,3\}\}$. Now $r = 2 \leq m$, we stop. The approximator is $\lceil\{1\}\rceil \vee \lceil\{2,3\}\rceil$.

Errors: The true OR outputs 0 only when none of $\{1,2\}, \{1,3\}, \{2,3\}$ appears, i.e., when vertices $1, 2, 3$ share the same color in a bipartite graph. There are 8 such colorings (all three same color, with the other two vertices colored arbitrarily). The approximator's singleton $\lceil\{1\}\rceil$ evaluates to 1 on every non-empty graph, so it wrongly accepts these 8 graphs, giving $E^- = 8$ and $E^+ = 0$. This demonstrates the OR rule: the approximator is $\geq$ the true function, only increasing E^-.

5.4 Composition of gates

Finally, consider an OR gate combining two approximated AND gates, e.g., $A = \hat{G}_{123} \vee \hat{G}_{124}$. Both AND approximations are constant 0, so their OR is 0. The true function $G_{123} \vee G_{124}$ is 1 on triangles $\{1,2,3\}$ and $\{1,2,4\}$. The approximator fails on both, yielding $E^+ = 2, E^- = 0$. The errors are simply inherited from the inputs; the OR step itself introduces no new errors. This verifies the separation of error types and the compositional nature of the approximation.

6 A Karchmer-Wigderson game for $\mathrm{CLIQUE}(3,5)$

The Karchmer-Wigderson (KW) duality [4] translates monotone circuit depth into the communication complexity of a search game. For $\mathrm{CLIQUE}(3,5)$, the game is defined as follows:

- Alice holds a 3-clique $S \subset \{1,2,3,4,5\}$ (one of the 10 triangles).
- Bob holds a 2-coloring $c : \{1,2,3,4,5\} \to \{0,1\}$ (representing a complete bipartite graph; edges exist only between vertices of different colors).
- Goal: find an edge $\{i,j\} \subseteq S$ such that $c(i) = c(j)$ (an edge present in Alice's clique but absent from Bob's bipartite graph).

By the pigeonhole principle, among the three vertices of S, at least two share the same color in c; hence a valid answer always exists.

6.1 An upper bound via a 4-bit protocol

Bob sends his full coloring in 4 bits: $c(2), c(3), c(4), c(5)$, fixing $c(1) = 0$ without loss of generality. Alice sends nothing. Alice, knowing her triangle $S = \{a,b,c\}$ and Bob's coloring c, checks the three edges of S. By the pigeonhole principle, at least one edge has both endpoints sharing the same color; she outputs the first such edge. Total communication: 4 bits. Hence $\mathrm{CC}(\mathrm{CLIQUE}(3,5)) \leq 4$.

6.2 A fooling set lower bound

We exhibit 2 input pairs (S_i, c_i) such that the unique correct answer is distinct and no single message can cover both:

1. $S_1 = \{1,2,3\}$, $\quad c_1$: $c(1) = c(2) = 0,\ c(3) = c(4) = c(5) = 1$ $\quad\Rightarrow$ unique answer $\{1,2\}$.
2. $S_2 = \{1,4,5\}$, $\quad c_2$: $c(4) = c(5) = 0,\ c(1) = c(2) = c(3) = 1$ $\quad\Rightarrow$ unique answer $\{4,5\}$.

For pair 1, the only monochromatic edge in S_1 under c_1 is $\{1,2\}$ (since $c(1)=c(2)=0$ while $c(3)=1$ causes the other two edges to be bichromatic). For pair 2, the only monochromatic edge in S_2 under c_2 is $\{4,5\}$ (since $c(4)=c(5)=0$ while $c(1)=1$).

Now consider whether a single protocol message could serve both pairs. If a message covers both (S_1,c_1) and (S_2,c_2), the protocol must output an edge that is simultaneously valid for both. But the answer sets are $\{\{1,2\}\}$ and $\{\{4,5\}\}$, which are disjoint. Hence no single message can cover both pairs. Therefore the two pairs form a fooling set of size 2, and

$$\mathrm{CC}(\mathrm{CLIQUE}(3,5)) \geq \lceil \log_2 2 \rceil = 1.$$

Combining the upper and lower bounds, we have

$$1 \;\leq\; \mathrm{CC}(\mathrm{CLIQUE}(3,5)) \;\leq\; 4,$$

which is consistent with the asymptotic KW lower bound $\Omega(\sqrt{k}) = \Omega(\sqrt{3}) \approx 1.73$.

7 Conclusion and open problems

We have shown that Razborov's exponential lower bound for monotone circuits computing CLIQUE remains essentially intact even if the circuit is allowed to use up to $c\log n$ negation gates. The proof is remarkably simple: the error caused by each NOT gate is at most $O(|X|+|Y|)$, while Razborov's threshold is $Q = |X|\cdot|Y|$; the huge ratio between Q and $|X|+|Y|$ leaves ample room for logarithmically many negations.

Compared to Amano and Maruoka [2], our proof handles a larger range of NOT gates ($c\log n$ vs. $(1/6)\log\log n$) by avoiding their delicate induction on circuit layers. The trade-off is that our error analysis is cruder: we replace each negated function by a constant approximator rather than constructing a precise approximator tailored to the function's structure. Since $|X|+|Y|$ is negligible compared to $Q = |X|\cdot|Y|$, this coarseness does not harm the asymptotic bound.

An obvious open problem is whether a similar statement holds for larger numbers of NOT gates, e.g., $s = O(n^{\varepsilon})$. The naive error analysis would require s to be $o(\min(|X|,|Y|))$, which is already superpolynomial in n, so the present method cannot be pushed much further without new ideas. A more refined approximator for negated functions (rather than a constant) might extend the range of s, at the cost of complicating the error analysis.

The hands-on examples (Sections 5 and 6) illustrate the core mechanisms in a concrete setting and are intended as pedagogical aids for researchers entering the field.

References

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